

Appendix II Standards for Plastics Testing

Standards for Plastics Testing				
Sr No.	Test Name	ASTM	ISO	BIS
A	Mechanical			
	Tensile Strength	ASTM D638 & D1708	--	IS 13360 (Part 5/Sec 1):1996
	Tensile Strength at Break	ASTM D638 & D1708	ISO 527-1, -2	
	Tensile Strength at Yield	ASTM D638 & D1708	ISO 527-1, -2	
	Tensile Strength, Ultimate	ASTM D638	ISO 527-1, -2	
	Tensile Modulus	ASTM D638	ISO 527-1, -2	
	Tensile Creep Modulus	--	ISO 527-1, -2	
	Tensile Elongation at Break	ASTM D638 & D1708	ISO 527-1, -2	
	Tensile Elongation at Yield	ASTM D638 & D1708	ISO 527-1, -2	
	Compressive Strength	ASTM D695	ISO 604	IS 13360(Part 5/Sec 8):1996
	Compressive Modulus	ASTM D695	ISO 604	
	Flexural Strength	ASTM D790	ISO 178	IS 13360(Part 5/Sec 6):1999
	Flexural Strength at Break	ASTM D790	--	
	Flexural Strength at Yield	ASTM D790	--	
	Flexural Modulus	ASTM D790	ISO 178	
	Flexural Creep	ASTM D2990	--	
	Flexural Creep Modulus	--	ISO 6602	
	Charpy Notched Impact Strength	ASTM D256	ISO 179	IS 13360(Part 5/Sec 5):1996
	Charpy Unnotched Impact Strength	--	ISO 179	
	Drop Impact Resistance	ASTM D4226	--	IS 13360(Part 5/Sec 24):1999
	Gardner Impact	ASTM D3029	--	
	Instrumented Dart Impact	ASTM D3763	ISO 7765-2	IS 13360 (Part 5/Sec 6):1999

	Multi-Axial Instrumented Impact Energy	--	ISO 6603-2	
	Multi-Axial Instrumented Impact Peak Force	--	ISO 6603-2	
	Notched Izod Impact Strength	ASTM D256	ISO 180	IS 13360(Part 5/Sec 4):1996
	Reverse Notch Izod Impact Strength	ASTM D256	ISO 180	
	Unnotched Izod Impact Strength	ASTM D256	ISO 180	
	Tensile Impact Strength	ASTM D1822	ISO 8256	
	Shear Modulus	ASTM D732	ISO 537	
	Shear Strength	ASTM D732	--	
	Apparent Bending Modulus	ASTM D747	--	
	Coefficient of Friction	ASTM D1894	--	IS 13360(Part 11/Sec 1):1999
	Deformation Under Load	ASTM D621	--	
	Fatigue Limit	ASTM D671	--	
	Poisson's Ratio	ASTM E132	--	
	Taber Abrasion Resistance	ASTM D1044	--	
	Hardness Hardness Conversion Chart	ASTM	ISO	
	Ball Indentation Hardness	--	ISO 2039-1	IS 13360(Part 5/Sec 12):1992
	Barcol Hardness	ASTM D2583	--	IS 13360(Part 5/Sec 14):2001
	Durometer (Shore) Hardness	ASTM D2240	ISO 868	IS 13360(Part 5/Sec 11):1992
	IRHD Hardness	ASTM D1415	ISO 48	
	Rockwell Hardness	ASTM D785	ISO 2039-2	IS 13360(Part 5/Sec 13):1992
B	Thermal			
	Accelerated Oven Ageing	--	ISO 4577	

	Brittleness Temperature	ASTM D746	ISO 812 & ISO 974	IS 13360(Part 6/Sec 11):2004
	DTUL (Deflection Temperature Under Load) at 1160 psi (8.0 MPa)	--	ISO 75	
	DTUL at 264 psi (1.80 MPa)	ASTM D648	ISO 75	IS 13360(Part 6/Sec 3):1997
	DTUL at 66 psi (0.45 MPa)	ASTM D648	ISO 75	
	Ductile/Brittle Transition Temperature	--	ISO 6603-2	IS 13360(Part 6/Sec 11):2004
	Glass Transition Temperature	ASTM E1356	--	
	Melting Temperature (DSC)	--	ISO 3146	IS 13360(Part 6/Sec 10):1992
	Peak Crystallization Temperature (DSC)	ASTM D3418	ISO 3146 & ISO 11357-1, -3	
	Specific Heat	ASTM C351	--	
	Thermal Conductivity	ASTM C177	ISO 8302	
	Vicat Softening Temperature	ASTM D1525	ISO 306	IS 13360(Part 6/Sec 1):1999
	Coefficient of linear Thermal Expansion	ASTM D696, ASTM E831, ASTM E228	ISO 11359-1, -2	IS 13360(Part 6/Sec 14):2004
C	Flammability			
	Flame Rating	--	--	
	Glow Wire Flammability Index	--	IEC 60695-2, -12	IS 13360(Part 6/Sec 5):2001
	Glow Wire Ignition Temperature	--	IEC 60695-2, -13	
	Limiting Oxygen Index	ASTM D2863	ISO 4589-1, -2	IS 13360(Part 6/Sec 6):2001
D	Electrical			
	Arc Resistance	ASTM D495	--	
	Comparative Tracking Index (CTI)	--	IEC 60112	IS 2824

	Dielectric Constant	ASTM D150 D2520	IEC 60250	
	Dielectric Strength	ASTM D149	IEC 60243-1	
	Dissipation Factor	ASTM D150 D2520	IEC 60250	
	Needle Flame Test	--	IEC 60695-11-5	
	Surface Resistivity	ASTM D257	IEC 60093	
	Volume Resistivity	ASTM D257	IEC 60093	IS 13360(Part 7/Sec 1):1996
E	Optical			
	Clarity	ASTM D1746	--	IS 13360(Part 9/Sec 8):2001
	Gardner Gloss	ASTM D523	--	
	Gloss	ASTM D2457	--	DOC.PCD 012(1678)
	Haze	ASTM D1003	--	IS 13360(Part 9/Sec 5):1999
	Refractive Index	ASTM D542	ISO 489	IS 13360(Part 9/Sec 1):2004
	Transmittance	ASTM D1003	--	IS 13360(Part 9/Sec 5):1999
	Yellowness	ASTM D1925	--	IS 13360(Part 9/Sec 9):2001
F	Physical			
	Apparent Density	ASTM D1895	ISO 60	IS 13360(Part 3/Sec 2):1997
	Bulk Factor	ASTM D1895	--	IS 13360(Part 3/Sec 4):1995
	Density -Specific Gravity	ASTMD1505, ASTMD792, & ASTMD4883	ISO 1183	
	Environmental Stress Cracking Resistance (ESCR)	ASTM D1693	--	IS 13360(Part 8/Sec 9):1997
	Melt Mass-Flow Rate (MFR)	ASTM D1238	ISO 1133	IS 13360(Part 4/Sec 1):1995
	Melt Volume-Flow Rate (MVR)	ASTM D1238	ISO 1133	IS 13360(Part 4/Sec 1):1995
	Molding Shrinkage	ASTM D955	ISO 294-4	
	PVC Cell Classification	ASTMD1784 & ASTMD2416	--	
	Vacuum Water Absorption	ASTM D1056		

	Viscosity Number	--	ISO 307	
	Water Absorption at 24 hrs	ASTM D570	ISO 62	
	Water Absorption at Equilibrium	ASTM D570	ISO 62	IS 13360(Part 8/Sec 1):1997
	Water Absorption at Saturation	ASTM D570	ISO 62	
G	Weathering Properties			
	Weathering by UV Lamp & Condensation		ISO 4892	
	Carbon Arc type light & water			
	Xenon Arc type light & water			DOC.PCD 012(1682)
H	Aging			
	Change in Durometer (Shore) Hardness	ASTM D471	ISO 1817	
	Change in Durometer (Shore) Hardness in Air	ASTM D573	ISO 188	
	Change in IRHD Hardness in Air	ASTM D573	ISO 188	
	Change in Tensile Strength	ASTM D471	ISO 1817	
	Change in Tensile Strength in Air	ASTM D573	ISO 188	
	Change in Ultimate Elongation	ASTM D471	ISO 1817	
	Change in Ultimate Elongation in Air	ASTM D573	ISO 188	
	Change in Volume	ASTM D471	ISO 1817	
	Change in Volume in Air	ASTM D573	ISO 188	